

LangChain Demo Document

LangChain is an open-source framework for building applications powered by large language models (LLMs). It provides tools for prompt management, chaining, retrieval, memory, agents, and integrations with external data sources.

Common use cases include chatbots, question-answering systems, Retrieval-Augmented Generation (RAG), document analysis, and AI agents that interact with APIs and databases.

A typical LangChain workflow involves loading documents, splitting them into chunks, generating embeddings, storing vectors in a vector database, retrieving relevant chunks, and using an LLM to generate answers.

LangChain RAG Example

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. Instead of relying solely on an LLM's training data, a RAG system retrieves relevant documents at query time.

Example Pipeline: PDF Documents → Text Chunks → Embeddings → Vector Database → Retriever → LLM → Answer with Citations

Benefits of RAG include reduced hallucinations, access to private knowledge bases, up-to-date information, and the ability to provide citations for generated answers.